

Agile Backlog Creation & Sprint Simulation

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Problem statement 16: Remote Patient Vitals Alert & Monitoring App. Post-operative patient telemetry includes SpO2, heart rate and blood pressure. The system validates incoming data, detects critical readings and alerts an on-call caregiver through an escalation matrix.

Jira workspace: Dhanush Patient Monitoring, project SCRUM1, company-managed Scrum board 3, hosted on the team member's Atlassian site. This individual backlog follows Dhanush's Lab 1 requirements.

Epic	Lab 1 requirements / responsibility
SCRUM1-1: Secure Patient Telemetry	FR-001: authenticated paired-device ingestion and timestamped readings.
SCRUM1-2: Telemetry Quality and Connection Monitoring	FR-002: validation and duplicate prevention; FR-003: disconnection and recovery.
SCRUM1-3: Critical Detection and Caregiver Escalation	FR-004: threshold evaluation, notification and ordered escalation.
SCRUM1-4: Authorized Dashboard and Alert History	FR-005: patient access restrictions, configuration, live view, acknowledgement and history.

Quality requirements: NFR-001 targets at least 500 concurrent streams and 99.99% availability. NFR-002 requires protected transmission/storage, authorization and audit records. These are design and future test targets; this exercise does not verify them.

Simulation scope: Four epics, ten stories and 44 estimated points. Two sprints were each configured for seven days and closed early after simulated To Do - In Progress - Done transitions. Jira Done denotes workflow demonstration, not software implementation. Updates were made through Vivaan's connected account.

Prioritized Backlog & Relative Estimates

Story	Summary	Priority	Points	Placement
SCRUM1-5	Accept telemetry only from authenticated paired devices	High	5	Sprint 1
SCRUM1-6	Store patient vital readings with measurement and receipt times	High	3	Sprint 1
SCRUM1-7	Validate telemetry and prevent duplicate records	High	5	Sprint 1
SCRUM1-8	Detect disconnected streams and restore monitoring on recovery	High	5	Sprint 1
SCRUM1-9	Evaluate configured clinical thresholds within two seconds	High	5	Sprint 2
SCRUM1-10	Notify caregivers and escalate unanswered critical alerts	High	8	Sprint 2
SCRUM1-11	Configure thresholds and restrict access to assigned patients	High	5	Sprint 1
SCRUM1-12	Display current vitals and acknowledge active alerts	High	3	Sprint 2
SCRUM1-13	Review and filter authorized patient alert history	Medium	3	Backlog
SCRUM1-14	Summarize telemetry rejection and connection events	Low	2	Backlog

Priority rationale: High-priority stories establish trusted measurements, secure access and emergency response. History filtering is Medium because immediate monitoring takes precedence. The audit summary is Low because source events are already logged; it improves review convenience.

Estimation: Fibonacci values 2, 3, 5 and 8 express relative effort and uncertainty, not hours. Estimates are provisional: a team Planning Poker discussion has not been confirmed. Review the acceptance criteria together, reveal votes, discuss differences and record agreed estimates before claiming this requirement complete.

User Stories: Telemetry Foundation

SCRUM1-5: Accept telemetry only from authenticated paired devices | 5 points | High

As a remote patient, I want to have my device authenticated and linked to my patient record, so that only my device can submit measurements for me.

Acceptance criteria: Accept valid paired device credentials; reject unknown or mismatched devices and log the reason without secrets; encrypt telemetry in transit.

Estimate rationale: Device authentication, pairing and access failure tests. **Traceability:** FR-001; NFR-002. **Parent:** SCRUM1-1.

SCRUM1-6: Store patient vital readings with measurement and receipt times | 3 points | High

As an on-call caregiver, I want to receive timestamped SpO2, heart rate and blood pressure, so that I can identify the patient and age of each measurement.

Acceptance criteria: Persist patient/device identity, measurement time, receipt time and units; associate each reading with the correct patient; preserve measurement time independently of receipt time.

Estimate rationale: Bounded ingestion and storage mapping. **Traceability:** FR-001. **Parent:** SCRUM1-1.

SCRUM1-7: Validate telemetry and prevent duplicate records | 5 points | High

As an on-call caregiver, I want to see only valid unique telemetry records, so that monitoring decisions are not distorted by corrupt or repeated data.

Acceptance criteria: Validate required identity, reading ID, times, values and units; reject incomplete records with a reason; replaying a reading ID creates no duplicate; extreme correctly formatted values still reach clinical evaluation.

Estimate rationale: Validation branches and idempotent storage tests. **Traceability:** FR-002. **Parent:** SCRUM1-2.

User Stories: Connection & Critical Response

SCRUM1-8: Detect disconnected streams and restore monitoring on recovery | 5 points | High

As an on-call caregiver, I want to see stale data and connection interruptions, so that I do not mistake old measurements for current readings.

Acceptance criteria: With a 30-second test inactivity timeout, mark Disconnected and label the last reading stale; keep its timestamp visible; next valid unique reading restores Connected; log loss and recovery.

Estimate rationale: Timers, stale-state handling and recovery logic. **Traceability:** FR-003. **Parent:** SCRUM1-2.

SCRUM1-9: Evaluate configured clinical thresholds within two seconds | 5 points | High

As an on-call caregiver, I want to have valid vitals evaluated promptly against patient thresholds, so that critical abnormalities are detected quickly.

Acceptance criteria: Evaluate valid readings within two seconds of receipt; a 145 BPM value breaches a configured 140 BPM upper limit; in-range values do not raise critical alerts; record patient, vital, value, threshold and detection time.

Estimate rationale: Time-sensitive evaluation and threshold boundary tests. **Traceability:** FR-004. **Parent:** SCRUM1-3.

SCRUM1-10: Notify caregivers and escalate unanswered critical alerts | 8 points | High

As a remote patient, I want to have critical alerts escalated when the first caregiver does not respond, so that another caregiver can receive the alert.

Acceptance criteria: Notify the assigned primary caregiver with patient/vital/value/time; use a configurable timeout (60 seconds in tests); escalate in matrix order; log attempts and exhaustion; stop pending escalation after acknowledgement.

Estimate rationale: Notification integration, ordered timers and acknowledgement races. **Traceability:** FR-004. **Parent:** SCRUM1-3.

User Stories: Authorized Caregiver Monitoring

SCRUM1-11: Configure thresholds and restrict access to assigned patients | 5 points | High

As an on-call caregiver, I want to configure thresholds only for my assigned patients, so that monitoring reflects authorized patient-specific settings.

Acceptance criteria: Require authentication; deny requests for unassigned patient IDs; validate threshold values; apply saved settings to subsequent evaluation; log caregiver identity and configuration changes.

Estimate rationale: Authorization and configuration validation. **Traceability:** FR-005; NFR-002. **Parent:** SCRUM1-4.

SCRUM1-12: Display current vitals and acknowledge active alerts | 3 points | High

As an on-call caregiver, I want to view current patient status and acknowledge an alert, so that the system records my response and stops further escalation.

Acceptance criteria: Display latest SpO2, heart rate, BP, measurement age and connection status; mark stale readings; acknowledgement records caregiver identity and time, sets Acknowledged, and cancels pending escalation; acknowledgement is not clinical resolution.

Estimate rationale: Dashboard integration using established access controls. **Traceability:** FR-005. **Parent:** SCRUM1-4.

User Stories: Remaining Backlog

SCRUM1-13: Review and filter authorized patient alert history | 3 points | Medium

As an on-call caregiver, I want to filter a patient's alert history, so that I can review prior monitoring events.

Acceptance criteria: Filter assigned patient history by date and state; show available detection, delivery, escalation and acknowledgement timestamps; provide an empty-state message; reject unauthorized history requests.

Estimate rationale: History queries and filtering. **Traceability:** FR-005. **Parent:** SCRUM1-4.

SCRUM1-14: Summarize telemetry rejection and connection events | 2 points | Low

As an on-call caregiver, I want to review telemetry quality events for assigned patients, so that I can identify recurring device transmission problems.

Acceptance criteria: List rejection reasons and disconnection/recovery events with timestamps; never expose credentials; enforce patient authorization; display no-events state when appropriate.

Estimate rationale: Simple filtered audit presentation. **Traceability:** FR-002/FR-003; NFR-002 (audit-view refinement). **Parent:** SCRUM1-2.

Backlog outcome: These two stories remain To Do (5 points). They were outside both sprint commitments and were not moved to Done. Consequently, the full product backlog and all Lab 1 functionality are not represented as complete.

Sprint Plan, Results & Reflections

Sprint	Committed scope	Result
Dhanush Sprint 1	SCRUM1-5, 6, 7, 8 and 11 23 points	5/5 stories; 23/23 points Done
Dhanush Sprint 2	SCRUM1-9, 10 and 12 16 points	3/3 stories; 16/16 points Done

Recorded times (IST): Sprint 1 planned 4 Oct 2026, 22:09 to 11 Oct, 22:09; closed 4 Oct, 22:13. Sprint 2 planned 4 Oct, 22:18 to 11 Oct, 22:18; closed 5 Oct, 11:20. The exercise resumed the next morning; elapsed wall-clock time is not development effort.

1. Did the estimates reflect actual effort?

They distinguish complexity: ordered escalation is 8 points because of timers, delivery integration and acknowledgement races; timestamp storage is 3. No implementation effort was measured, so estimation accuracy cannot be established from this simulation.

2. Was the backlog well-prioritized?

Sprint 1 established authentication, valid unique data, connection state and authorized configuration. Sprint 2 used those foundations for critical detection and caregiver response. Lower-priority history and audit views remain outside the commitments.

3. How did the simulated sprints align with the plan?

All eight committed stories passed through In Progress before Done. Sprint 1 completed 23 points and Sprint 2 completed 16, without scope additions or removals. Both retained seven-day planned durations but closed early during the workflow exercise.

4. What did the burndown reveal about capacity?

Remaining points reached zero in both sprints. Sharp drops show when statuses changed, not a measured delivery rate. The overnight pause in Sprint 2 explains its longer flat segment. The 39 completed points cannot be used as a reliable weekly capacity forecast; real implementation sprints are needed.

Completion checks: Backlog, active boards and final burndowns are documented below. Team Planning Poker confirmation and the instructor workspace demonstration remain to be completed/confirmed.

Backlog, Epics & Estimates

The screenshot displays the Jira interface for the SCRUM1 board. The left sidebar shows the navigation menu with options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Recent', 'Recommended', 'Filters', 'Dashboards', 'Assets', 'Teams', 'Goals', 'Projects', and 'Customise sidebar'. The main content area is titled 'SCRUM1 board' and shows the 'Backlog' view. The 'Epic' section on the left lists four epics: 'No epic', 'Secure Patient Telemetry', 'Telemetry Quality and Connection Monitoring', and 'Critical Detection and Caregiver Escalation'. The 'Sprints' section on the right shows two sprints: 'Dhanush Sprint 1' (5 work items, 23 points) and 'Dhanush Sprint 2' (3 work items, 16 points). The 'Backlog' section at the bottom shows two additional stories: 'SCRUM1-5' and 'SCRUM1-6'.

Sprint	Item	Key	Description	Assignee	Points	Status
Dhanush Sprint 1 (23 points)	SCRUM1-5	Accept telemetry only from authenticated paired devices	Secure Patient Tel...	To Do	5	Not Started
	SCRUM1-6	Store patient vital readings with measurement and receipt times	Secure Patient Tel...	To Do	3	Not Started
	SCRUM1-7	Validate telemetry and prevent duplicate records	Telemetry Quality ...	To Do	5	Not Started
	SCRUM1-8	Detect disconnected streams and restore monitoring on recovery	Telemetry Quality ...	To Do	5	Not Started
	SCRUM1-11	Configure thresholds and restrict access to assigned patients	Authorized Dashb...	To Do	5	Not Started
Dhanush Sprint 2 (16 points)	SCRUM1-9	Evaluate configured clinical thresholds within two seconds	Critical Detection ...	To Do	5	Not Started
	SCRUM1-10	Notify caregivers and escalate unanswered critical alerts	Critical Detection ...	To Do	8	Not Started
	SCRUM1-12	Display current vitals and acknowledge active alerts	Authorized Dashb...	To Do	3	Not Started
Backlog (2 work items)	SCRUM1-5	Accept telemetry only from authenticated paired devices	Secure Patient Tel...	To Do	5	Not Started
	SCRUM1-6	Store patient vital readings with measurement and receipt times	Secure Patient Tel...	To Do	3	Not Started

Four epics and both future sprints before execution: Sprint 1 = 23 points; Sprint 2 = 16 points. Two additional stories remain in the backlog.

Sprint 1: Workflow Progress

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Filters Dashboards Assets Teams Goals Projects

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SCRUM1 board

Summary Timeline **Active sprints** Backlog Calendar List Forms Development Docs +

Complete sprint Group: Story

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TO DO

+ Create

IN PROGRESS 3

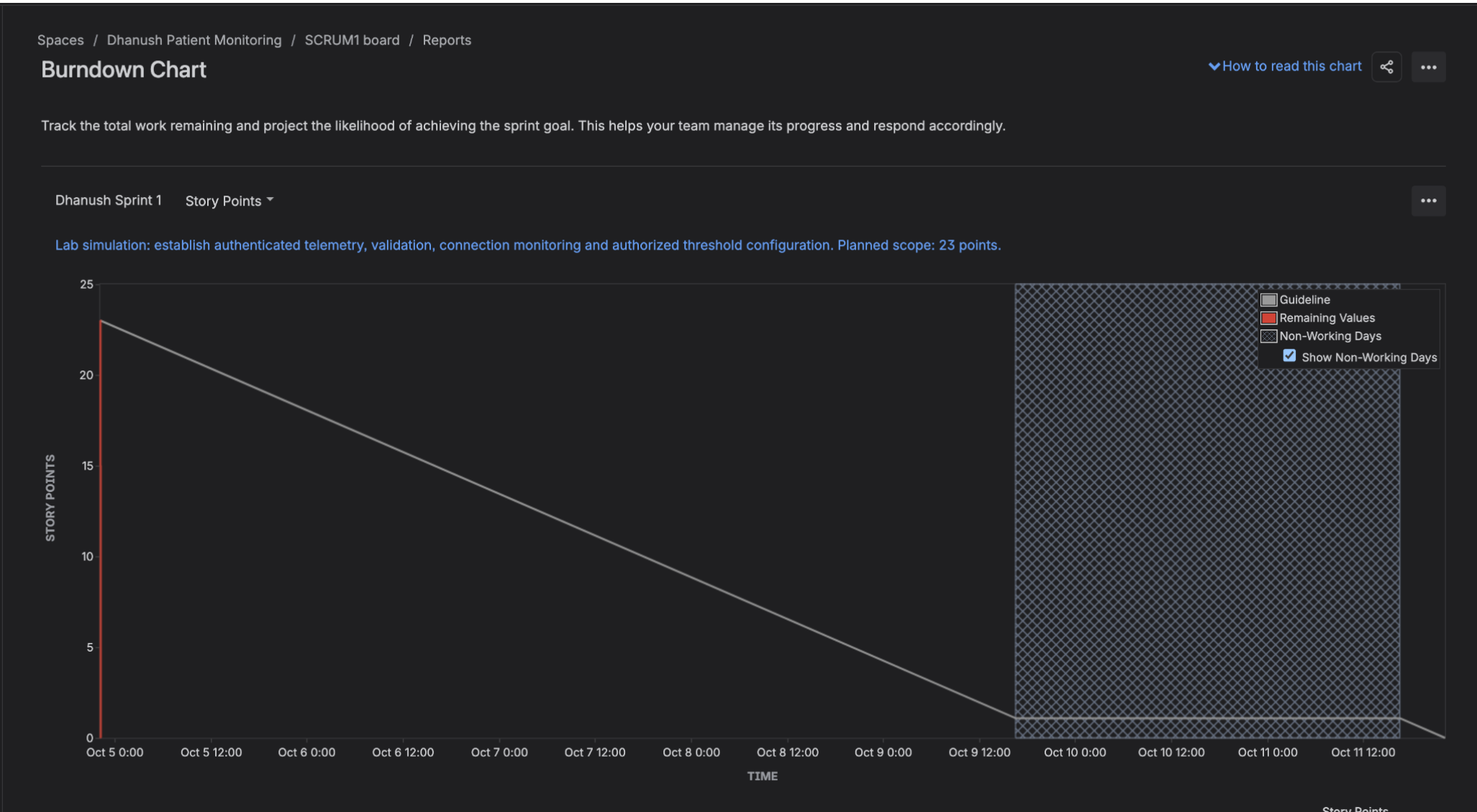
- Validate telemetry and prevent duplicate records
 - Telemetry Quality and Connection Monitoring
 - ✓ SCRUM1-7 5
- Detect disconnected streams and restore monitoring on recovery
 - Telemetry Quality and Connection Monitoring
 - ✓ SCRUM1-8 5
- Configure thresholds and restrict access to assigned patients
 - Authorized Dashboard and Alert History
 - ✓ SCRUM1-11 5

DONE 2

- Accept telemetry only from authenticated paired devices
 - Secure Patient Telemetry
 - ✓ SCRUM1-5 5
- Store patient vital readings with measurement and receipt times
 - Secure Patient Telemetry
 - ✓ SCRUM1-6 3

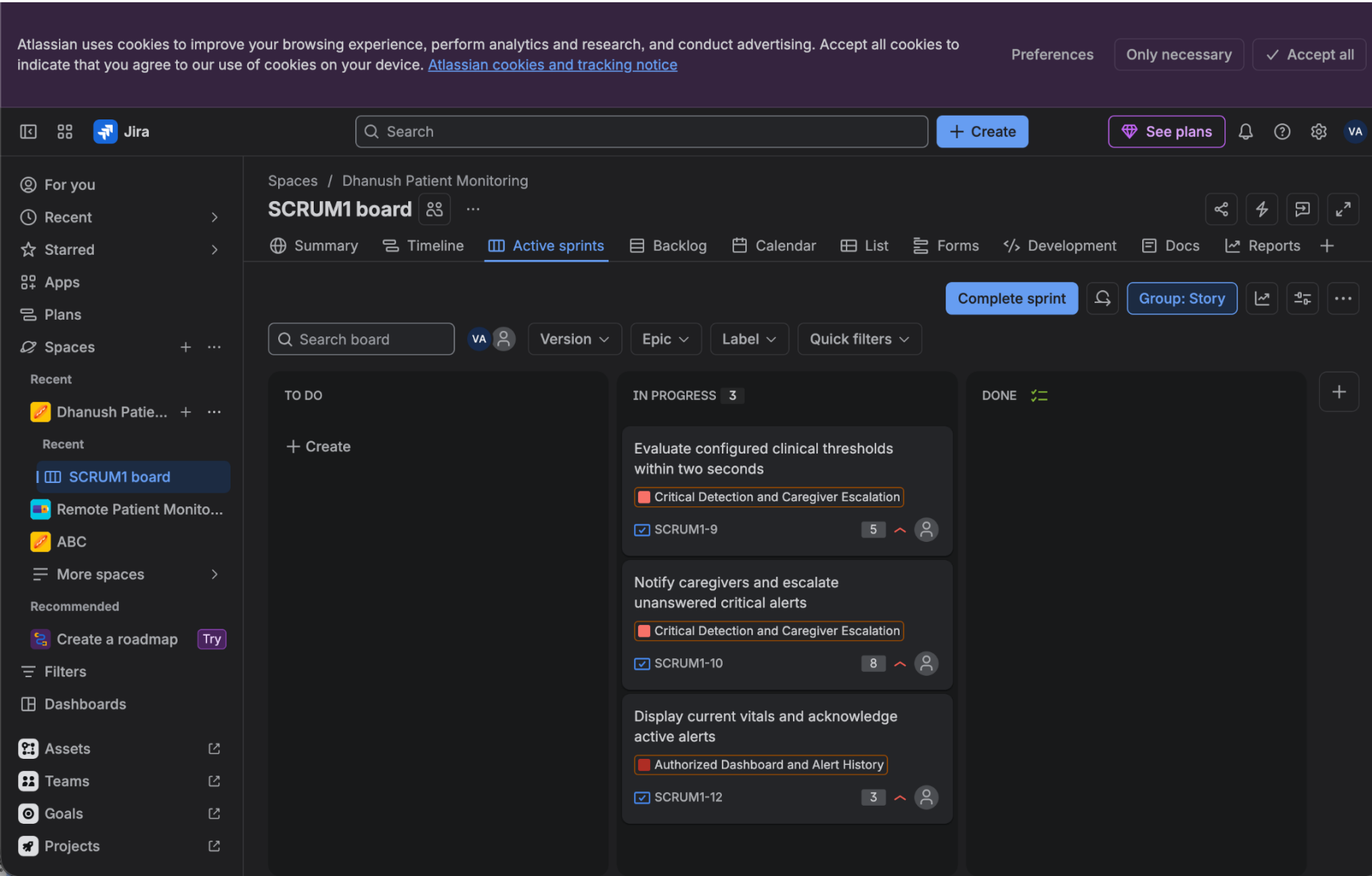
Two stories Done (8 points) and three In Progress (15 points). All five committed stories subsequently reached Done.

Sprint 1: Final Burndown



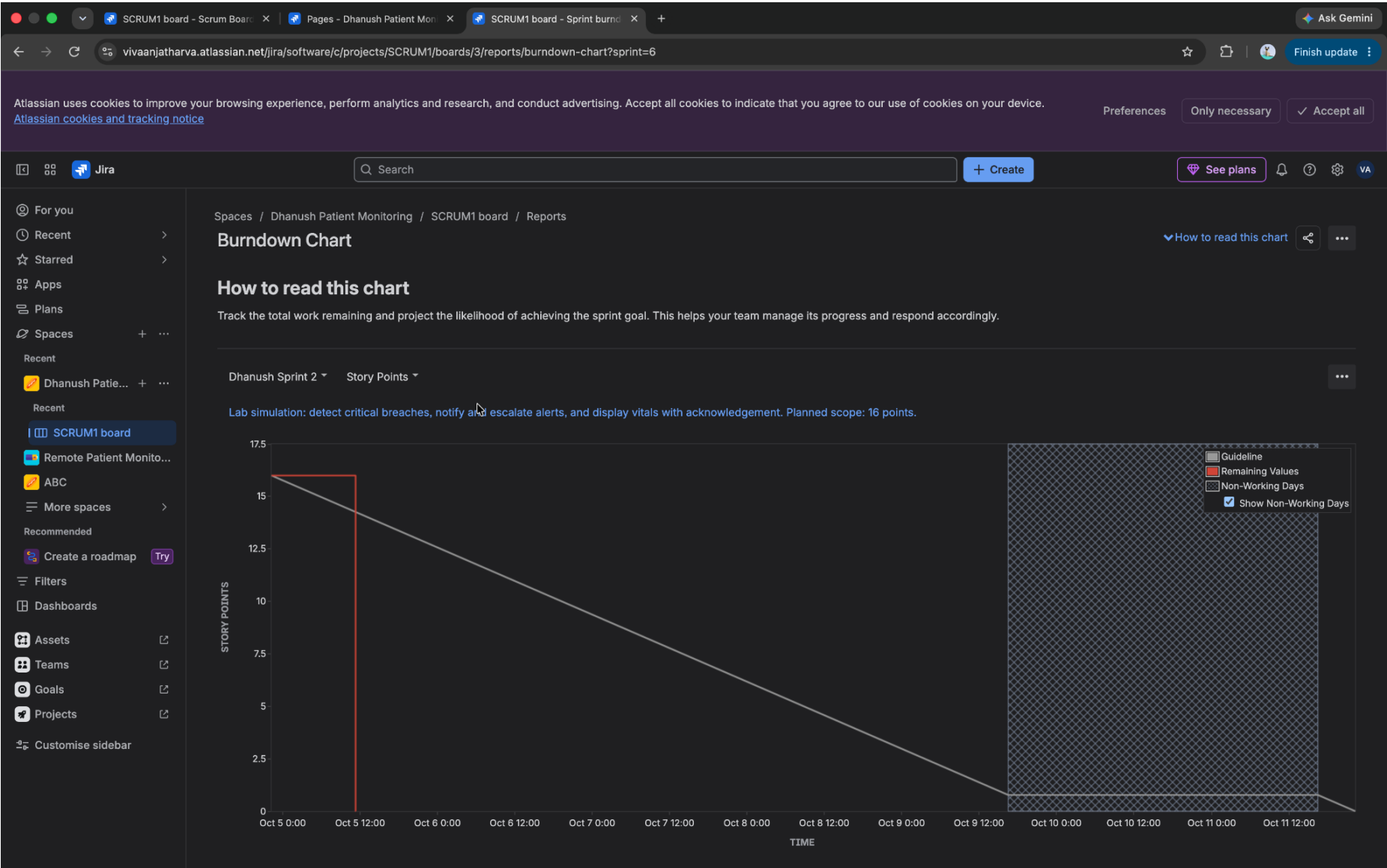
Remaining work falls from 23 points to zero. The seven-day guideline is retained; the early drop reflects compressed simulation.

Sprint 2: Active Board



SCRUM1-9 (5), SCRUM1-10 (8) and SCRUM1-12 (3) are In Progress before completion: 16 points in total.

Sprint 2: Final Burndown



Remaining work falls from 16 points to zero on 5 October. The sprint was closed after the simulated transitions; its planned end remains 11 October.